Introduction to Data Analytics
MS - 4610

Session 7 – EVPI, EVSI, conjugate priors

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Expected Value of Perfect Information (EVPI)

- EVPI is the expected price that we are willing to pay to gain access to perfect information.

- EVPI = Expected value with perfect information - Expected value without perfect information

Expected value of sample information (EVSI)

- EVSI is the expected increase in monetary value by obtaining access to sample of additional observations.

- EVSI = Expected value with sample information – Expected value without sample information

Example 1

- Consider a situation in which there is 80% chance that stock price goes up, and 20% chance that it goes down. The profit we make by investing when price is up is Rs.10, and the loss we incur by investing when price is down is Rs.20. Calculate EVPI

- The payoff matrix can be represented as follows:

Decision	Stock up	Stock down
Buy	10	-20
Not buy	0	0
Probability	0.8	0.2

Example 1

- Expected value without perfect information:
  - Expected monetary value = (10*0.8) + (-20*0.2)

    (EMV) = 4
- Expected value with perfect information (EV|PI):
  - If it is know that the stock is up, we make a decision to buy. If it is down, we make a decision to not buy.
  - EV|PI = (0.8*10) + (0.2*0) = 8
- EVPI = EV|PI – EMV

  = 8 - 4 = 4

Example 1: EVSI scenario 1

- Considering Example 1, assume we have the stock price status for previous 5 years. From this sample of 5 years, 90% of time we predict the current status correctly. 10% of the time the prediction is wrong.

- That is, if the stock is actually up, we predict that it is up with probability 0.9, and predict that it is down with probability 0.1

Example 1: EVSI scenario 1

Decision	Stock up	Stock down
Buy	10	-20
Not buy	0	0
Probability	0.8	0.2

Actual status

Sample info

	Up	Down
Up	0.9	0.1
Down	0.1	0.9

- EV|SI = (0.8*0.9*10) + (0.1*0.2*(-20)) = 6.8
- We know that, EMV = 4
- EVSI = EV|SI – EMV

  = 6.8 – 4 = 2.8
- Note: EVPI value was 4

Example 1: EVSI scenario 2

- Suppose for example 1, the probabilities from the sample is modified by the given matrix.

Actual status

Sample info

	Up	Down
Up	0.7	0.2
Down	0.3	0.8

- EV|SI = (0.8*0.7*10) + (0.2*0.2*(-20)) = 4.8
- We know that, EMV = 4
- EVSI = EV|SI – EMV

  = 4.8 – 4 = 0.8

Decision	Stock up	Stock down
Buy	10	-20
Not buy	0	0
Probability	0.8	0.2

General formulae

- Revision of Probability:
  - PDF: f(x)
  - It's relation to CDF: F(x)
  - Mean: E(f(x))
  - Functions of random variables: dealing with f(x) and g(x)
- EMV in general
  - f(x) is the PDF of possible outcomes
  - g(x) is the function of the objective variable that we want to maximize or minimize
  - i is the decision variable

$$\max_i \int_{-\infty}^{\infty} f(x).\, g_i(x).\, dx$$

General formulae

- EVPI

$$\int_{-\infty}^{\infty} f(x).\max_i(g_i(x)).dx$$

- EVSI
  - We need a functional mapping between truth ‘x’ and your estimate of truth u(x)
  - Class exercise.

Conjugate priors

- We can make distributional assumptions on the parameters of a distribution.
- We can use Bayes theorem to update our assumptions following a sample from a likelihood function
- We are dealing with 2 distributions
- When the posterior distribution is in the same family as the prior then they are called conjugate distributions

Conjugate priors

- Using Bayes' theorem

$$p(\theta \mid x) = \frac{p(x \mid \theta)\, p(\theta)}{p(x)}$$

$$= \frac{p(x \mid \theta)\, p(\theta)}{\int p(x \mid \theta')\, p(\theta')\, d\theta'}$$

Likelihood	Model parameters	Conjugate prior distribution	Prior hyperparameters	Posterior hyperparameters	Interpretation of hyperparameters
Normal with known variance σ^2	μ (mean)	Normal	$\mu_0,\ \sigma_0^2$	$\frac{1}{\frac{1}{\sigma_0^2}+\frac{n}{\sigma^2}}\left(\frac{\mu_0}{\sigma_0^2}+\frac{\sum_{i=1}^n x_i}{\sigma^2}\right),\ \left(\frac{1}{\sigma_0^2}+\frac{n}{\sigma^2}\right)^{-1}$	mean was estimated from observations with total precision (sum of all individual precisions) $1/\sigma_0^2$ and with sample mean μ_0

Wikipedia contributors. "Conjugate prior." *Wikipedia, The Free Encyclopedia*. Wikipedia, The Free Encyclopedia, 5 Sep. 2019. Web. 17 Sep. 2019.